

Internal Assessment Automation Web Application

Project Specification Document

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1. Introduction

1.1 Purpose of this Document

This document specifies the SEBSA Internal Assessment Automation Web Application: what it does, who uses it, its functional and non-functional requirements, its system architecture, database design, hosting plan, indicative cost, and a suggested build roadmap. It is intended as the single reference for whoever designs, builds, and signs off on this project.

1.2 Background

SEBSA currently needs a recurring, twice-yearly internal knowledge assessment process for employees. Today this is a manual effort: distributing a test, collecting responses, and issuing proof of completion. This project replaces that manual process with a small internal web application that HR/Admin can operate without developer involvement once built — managing a question bank, triggering the assessment email, and tracking completion, while employees get an instant certificate on submission.

1.3 Document Scope

This is a specification and planning document. Technology choices are recommendations sized for SEBSA's stated scale (50–200 employees) and can be revised by whoever implements the project. Cost figures are indicative only — see Section 12.

2. Project Overview & Objectives

The system automates SEBSA's internal assessment cycle end-to-end: HR/Admin maintains a bank of 60 questions; every assessment cycle (nominally every 6 months) each employee is presented with 10 questions drawn at random from that bank, in shuffled order; on submission the system scores the attempt and immediately generates a downloadable certificate showing the employee's score; and HR/Admin can see, at a glance, who has completed the current cycle and who has not.

2.1 Objectives

- Remove the manual effort of distributing the assessment and chasing completions every 6 months.
- Give employees an immediate, self-service certificate the moment they finish.
- Give HR/Admin one screen to manage questions, one action to launch a cycle, and one screen to track completion and scores.

3. Scope

3.1 In Scope (Phase 1 / MVP)

- Employee login using company-issued credentials (email + password).
- Randomised selection of 10 questions from an active bank of up to 60, with shuffled question and answer order.
- Single-page assessment flow: answer all 10, submit once.
- Automatic scoring and instant certificate (PDF) generation and download on submission, showing the employee's score.
- Admin/HR Question Bank management (add, edit, activate/deactivate questions).
- Admin/HR Employee list management (add, edit, activate/deactivate employees for login and email dispatch).
- One-click “Send Assessment” action that opens a new cycle and emails the assessment link to all active employees.
- Admin/HR completion tracker: who has completed, who has not, scores, and completion dates for the active cycle.

3.2 Out of Scope (Phase 1)

Flagged here so they are not assumed to be included unless explicitly added to Phase 1:

- Integration with any existing SEBSA HR/payroll system or company single sign-on (SSO) / Active Directory - login is a standalone email + password in v1.
- Employee self-registration - accounts are created and maintained by HR/Admin.
- Automated reminder emails to employees who have not yet completed the assessment (listed as a Phase 2 enhancement, Section 13).

4. Assumptions & Dependencies

The requirements described to build this document did not specify every detail. The points below are assumptions made to produce a complete, buildable spec — please review and correct any that don't match what SEBSA actually needs before development starts.

- Questions are multiple-choice with exactly one correct option each.
- One assessment cycle is active company-wide at a time; all active employees receive the same cycle and question bank.
- The employee list (assumed in the 50–200 range) is entered/maintained inside the tool itself, not pulled live from another HR system.
- The application UI is in English.
- **No pass/fail threshold is enforced: every submitted attempt is scored and a certificate is issued showing that score, as confirmed for this document.**
- Employees access the system via company login (email + password) rather than a one-time personalised link, as confirmed for this document.

5. User Roles & Permissions

Role	Description	Key Permissions
Employee	Any SEBSA staff member being assessed.	Log in Take one assessment per active cycle View and download own certificate(s)
HR / Admin	HR staff operating the tool.	Manage question bank Manage employee list Start a cycle / send assessment email View tracker & scores

Note: A separate "Super Admin" role (e.g. to manage other Admin accounts) is not included in Phase 1 but can be added later if SEBSA wants to separate HR operators from system owners.

6. Functional Requirements

6.1 Employee-Facing

ID	Requirement	Description	Priority
FR-01	Login	Employee logs in with company email + password.	Must
FR-02	Cycle access	On login during an active, not-yet-attempted cycle, the employee is taken to the assessment.	Must
FR-03	Random selection	System randomly selects 10 questions from the active question bank (up to 60) for this attempt.	Must
FR-04	Shuffle	Selected questions and their answer options are shown in a randomised order, different per employee.	Must
FR-05	Answer & submit	Employee answers all 10 questions and submits in a single action.	Must
FR-06	Instant scoring	System auto-scores the submission immediately.	Must
FR-07	Certificate generation	On submit, a PDF certificate is generated showing employee name, cycle name, score, date, and a unique certificate ID.	Must
FR-08	Certificate download	On successful submission, the system generates the certificate PDF, emails it to the employee's registered email address with the designated HR/Admin email address in CC, and provides an immediate download option.	Must
FR-09	Certificate history	Employees can log back in later and re-download certificates from past cycles.	Should
FR-10	**One attempt per cycle**	**Employees cannot retake the assessment once submitted, within the same cycle.**	Must

6.2 Admin / HR-Facing

ID	Requirement	Description	Priority
FR-11	Question bank management	Add, edit and activate/deactivate questions; each has question text, answer options, and the correct answer marked. Target bank size: 60.	Must
FR-12	Employee management	Add, edit and activate/deactivate employee records (name, email, department) used for login and email dispatch.	Must
FR-13	Start cycle / send assessment	One action opens a new assessment cycle and emails the assessment link to all active employees.	Must

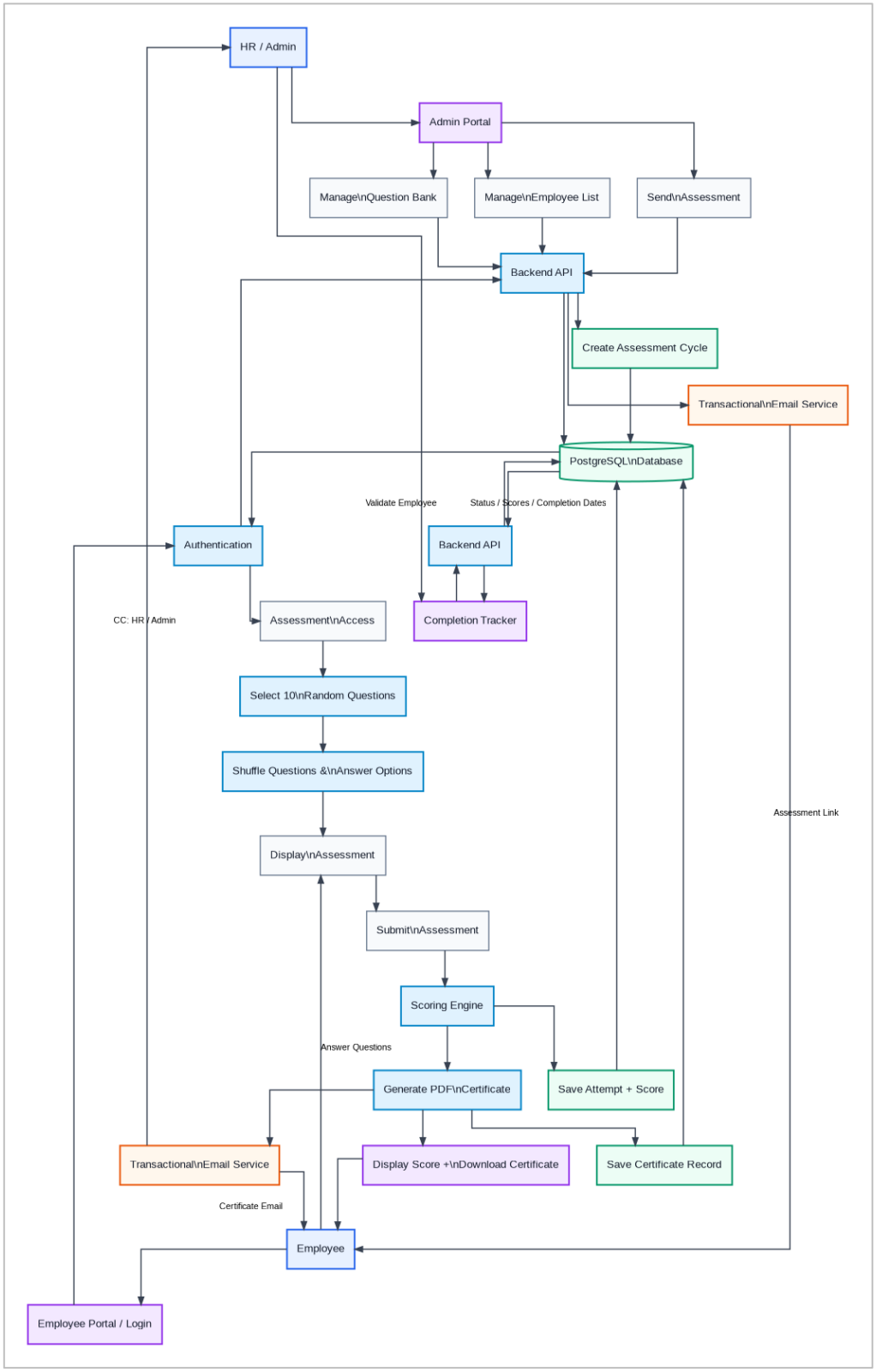
ID	Requirement	Description	Priority
FR-14	Selective resend	Resend the link to specific employees (e.g. those who missed it) without re-emailing everyone.	Should
FR-15	Completion tracker	Dashboard listing employees for the current cycle with status (Not Started / Completed), score, and completion date.	Must
FR-16	Historical reports	View and filter tracker data from previous cycles.	Should
FR-17	Certificate re-issue	Admin can re-download or re-send an employee's certificate.	Should
FR-18s	Export tracker	Export the tracker table (e.g. CSV/Excel) for HR records.	Could

7. Non-Functional Requirements

Category	Requirement
Usability	Simple, clean, modern interface usable without training, on both desktop and mobile browsers.
Performance	Assessment screens should feel instant in normal use; the system should comfortably handle the full 50–200 employee base attempting within the same week without slowdown.
Availability	Reasonable uptime for an internal HR tool (e.g. targeting ~99% during business hours is a sensible planning goal); brief maintenance windows are acceptable outside an active cycle.
Security	Passwords stored hashed, never in plain text; all traffic over HTTPS; role-based access control between Employee and Admin; sessions expire after inactivity.
Data integrity	Once submitted, an attempt's answers, score and certificate are recorded immutably — no silent re-editing after submission.
Scalability	Architecture should support growth beyond 200 employees without a redesign, even though it is being sized for 50–200 today.
Maintainability	HR/Admin can manage the question bank and employee list entirely through the UI — no code changes needed for routine operation.
Auditability	The system records when each cycle was opened/sent and by which Admin.
Data privacy	Assessment records are internal HR data, restricted to the Admin role; see Section 11 for a recommendation on retention and compliance review.
Browser support	Current versions of Chrome, Edge, Safari and Firefox, desktop and mobile.

8. System Architecture

The diagram below shows the recommended high-level architecture: a browser-based frontend for both employees and Admin/HR, a backend API that holds the assessment, question-bank, certificate and email logic, a relational database, and a transactional email provider for the 6-monthly dispatch — all running on a low-cost cloud hosting setup (Section 12).



8.1 Request Flow (Employee)

- Employee opens the emailed link → logs in with company credentials.
- Backend confirms an active cycle and that this employee has not already attempted it, then picks 10 random questions from the active bank and shuffles them and their options.
- Employee answers and submits → backend scores the attempt, generates the certificate PDF, and stores the attempt + certificate record.
- Employee is shown the score and a download button for the certificate immediately.

8.2 Request Flow (Admin/HR)

- Admin manages the question bank and employee list directly through the Admin Panel.
- Admin clicks “Send Assessment” → backend opens a new cycle and calls the email service to dispatch the link to every active employee.
- Tracker screen queries attempt records in real time to show who has completed the current cycle.

9. Recommended Technology Stack

These are recommendations sized for a small internal tool at SEBSA's stated scale — a developer can substitute equivalents without changing the requirements above.

Layer	Recommendation	Why
Frontend	React (or Next.js) with a component library / Tailwind CSS	Matches the “simple, modern, clean” UI goal; large ecosystem, easy to hand over to another developer later.
Backend / API	Node.js with Express, or Next.js API routes if Next.js is used for the frontend	One language (JavaScript/TypeScript) across the stack; Next.js can host frontend + backend as a single deployable app, which simplifies hosting.
Database	Managed PostgreSQL (e.g. Supabase, Neon, or the hosting platform's managed Postgres)	Relational structure fits Employees / Questions / Attempts / Certificates cleanly; managed hosting removes backup/patch overhead.
Authentication	Email + password, hashed with bcrypt/argon2, signed session tokens (JWT or server sessions)	Matches the confirmed “company login” requirement; SSO/AD integration can be added later without a rebuild.
Certificate generation	A server-side PDF library (e.g. pdf-lib) or HTML-to-PDF rendering (e.g. Puppeteer)	Produces a consistent, branded PDF on submit.

Layer	Recommendation	Why
Transactional email	Amazon SES, SendGrid, or Resend	Reliable delivery for the twice-yearly bulk send; supports domain authentication (SPF/DKIM) to avoid spam folders.
File storage	Cloud object storage (S3-compatible), or storing PDFs in the database at this scale	Either is workable for 50–200 employees; object storage is the more conventional choice if the team expects to grow.
Hosting	A low-cost PaaS (Render, Railway, Fly.io) or a small VPS (DigitalOcean, Linode) + managed database	Sized for a low-traffic internal tool — see Section 12 for cost.

Note: I'm not fully certain these are still each provider's current recommended tier or pricing plan — please verify on the provider's site before committing, since these details change over time.

10. Database Design (High Level)

This is an indicative schema to guide development, not a final DDL — the developer will likely refine field names and add indexes/constraints.

Entity	Key Fields	Notes
Employees	id, full_name, email (unique), department, status, password_hash, created_at	One record per SEBSA employee who can log in.
Questions	id, question_text, option_a–d, correct_option, category, status, created_at	The question bank; status lets Admin retire a question without deleting history.
AssessmentCycles	id, label (e.g. “2026 H2”), opened_at, opened_by, status	One record per 6-monthly cycle.
Attempts	id, employee_id (FK), cycle_id (FK), selected_question_ids, answers, score, submitted_at	One record per employee per cycle; the 10 selected questions and answers are stored for audit/record purposes.
Certificates	id, attempt_id (FK), certificate_number (unique), file_path, issued_at	One record per issued certificate; certificate_number supports later verification if needed.
AdminUsers	id, full_name, email, password_hash, role	HR/Admin accounts, separate from Employees.

11. Assessment Cycle & Email Workflow

- HR/Admin clicks “Send Assessment” (manually, roughly every 6 months) — an automatic calendar trigger can be added later if preferred, but a manual trigger keeps HR in control of exact timing.
- The system opens a new AssessmentCycle and records who it was sent by and when.
- A bulk email goes out, via the transactional email provider, to every active employee's registered address. Because login is by company credentials rather than a one-time token, the email links to the login/assessment page — the system identifies the employee once they sign in.
- Each employee can log in at any time before the cycle closes and complete the assessment once.
- The tracker updates in real time as employees complete their attempt.
- A configurable completion window (e.g. 2 weeks) is recommended so HR can plan follow-ups; the cycle can be closed manually or left open at HR's discretion in v1.

12. Certificate Design & Content

Each certificate (A4 PDF) should contain:

- Employee full name
- Assessment/cycle name (e.g. “SEBSA Internal Assessment — 2026 H2”)
- Score achieved (e.g. “8 / 10”), as confirmed — shown regardless of whether a pass/fail threshold is used
- Date completed
- A unique certificate ID/number
- SEBSA branding (logo and brand colours — see the accompanying UI direction)

The certificate is available for download immediately after submission, and can be re-downloaded later from the employee's own login and from the Admin tracker (FR-09, FR-18).

13. Security Considerations

- Passwords hashed with a strong algorithm (bcrypt or argon2) — never stored or logged in plain text.
- All traffic served over HTTPS/TLS.
- Role-based access control strictly separating Employee and Admin capabilities.
- Login rate-limiting/lockout to reduce brute-force risk.
- Server-side input validation on all forms (login, question bank entries) to guard against injection.
- Admin actions (question edits, cycle sends, employee changes) are logged for auditability.
- Regular automated database backups, with a documented restore process.

.14. Hosting Plan & Indicative Cost

Component	Suggested option	Indicative monthly cost (USD)
App hosting	Small PaaS instance (Render/Railway) or a 1–2GB VPS (DigitalOcean/Linode)	I believe roughly $5–$25, depending on provider/tier — please verify
Managed database	Small managed PostgreSQL instance (Supabase/Neon/hosting platform's own)	I believe roughly $0 (limited free tier) to $20–$25 — please verify
Transactional email	Amazon SES, SendGrid, or Resend	At ~50–200 emails twice a year, this is very low volume — likely free or the lowest paid tier (a few USD/month); please verify current free-tier limits
File/object storage	S3-compatible bucket for certificate PDFs (optional — DB storage may suffice at this scale)	Typically cents to a couple of USD/month at this volume
Domain name	A subdomain of sebsaworld.com, or a new domain if preferred	Roughly $10–$15/year if a new domain is purchased — please verify with a registrar
SSL/HTTPS	Let's Encrypt or platform-included certificate	Usually free

15. Development Phases / Roadmap

15.1 Phase 1 — MVP (this specification)

Everything listed in Section 3.1: login, random+shuffled assessment, instant scored certificate, Admin question bank, employee list, send-assessment action, and completion tracker.

15.2 Phase 2 — Suggested Future Enhancements

- Automated reminder emails to employees who haven't completed before the window closes.
- Single sign-on / company directory integration for login.
- CSV/Excel export of the tracker (if not already built in Phase 1 as FR-19).
- Support for additional question types beyond multiple choice.
- An optional configurable pass/fail threshold, if SEBSA later wants one.
- Sinhala/English language toggle.
- Basic analytics (average scores by department, trend across cycles).

16. Risks & Mitigations

Risk	Mitigation
Employees share answers/questions with colleagues yet to attempt.	Randomised 10-of-60 selection with shuffled order reduces, but cannot fully eliminate, this — refresh/expand the question bank periodically.
Employees forget login credentials.	Provide a self-service "forgot password" reset flow.
Assessment email lands in spam / low deliverability.	Use a reputable transactional email provider with proper domain authentication (SPF/DKIM/DMARC).
Single low-cost server is a single point of failure.	Acceptable for an internal tool at this scale; mitigate with regular automated backups and a documented, tested restore process.
Scope creep during build.	Treat this document's Section 3.1 as the Phase 1 contract; route later requests through Section 15.2.

17. Appendix

A. Sample Question Record

Field	Example value
question_text	"Which of the following is the strongest example of a phishing indicator?"
option_a	"An email from a known colleague with no attachment"
option_b	"An urgent request to click a link and re-enter your password"
option_c	"A calendar invite from your manager"
option_d	"A newsletter you subscribed to"
correct_option	b
category	Security Awareness (example — actual categories depend on SEBSA's real question set)

Note: This is an illustrative example only to show the record shape — the actual 60 questions and their subject matter are to be supplied by SEBSA.

B. Glossary

Term	Meaning
Cycle	One instance of the twice-yearly assessment period, opened by Admin and sent to all active employees.

Term	Meaning
Attempt	One employee's single completed run of an assessment within a cycle.
Question bank	The full set of up to 60 maintained questions Admin manages; 10 are drawn per attempt.
Certificate	The PDF generated on submission, showing the employee's score for that cycle.